



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: I Semester B.Tech.IT

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs. G. Mareeswari

Innovative Practice Description

Unit/Topic: I/ Illustrative problem: Insert a card in a list of sorted cards

Course Outcome: CO 1

Topic Learning Outcome: TLO1, 2, 3

Activity Chosen: Drawing for Understanding.

Date of Implementation: 09.10.2023

Justification:

- Drawing for understanding give students the opportunity to understand the concept by formulating the pictures and get familiar in the topic. It forces students to grapple with what they're learning and reconstruct it in a way that makes sense to them. The researchers also suggest that drawing results in better recall because of how the information is encoded in memory. This method brings life to the subject matter, stimulates engagement from passive learners, and helps them retain it. Insert a card in a list of sorted cards is similar to the kind of playing cards game where the students are involved to play the game as per the rule.
- **Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes.
- All the students in the class actively involved in this activity.
- Students were asked to draw the empty cards in their paper. Instructor dictate the numbers and the students write the list of numbers one by one in each card and tear the cards and hold it on the table.
- Students start with a note that the cards face down on the table.
- Then remove one card at a time from the table and place it into the correct position in the left hand.
- To find a correct position for a card, they compare it with each of the cards already in the hand from left to right.
- Once the position is found, the cards from that position are moved to the next higher indexed position and in that order.

- The final card is inserted at the current position
- After all the cards are sorted, the instructor mentioned another number and students follow the same procedure to insert the card in the particular position
- The objective of this practice is to insert the card in to the sorted list of cards by comparing the values stored from left to right. For this they follow the below mentioned rules
 - Initially the new card is inserted to the left and compare with the first value.
 - If the value in the first card is greater than the value of new card, the new card is fixed at the zeroth index.
 - Otherwise swap the first card and new card, then new card is compared with the next card, if new card value is less than next card then new card is inserted at the current position otherwise again swapping takes place.
 - Similar to this the same process is repeated until the new card reaches its correct position.
- Students shown large interest while doing this activity and they successfully completed the activity in their first attempt.
- This activity was very helpful to the students in recalling the topic taught on that specific day, generating new ideas about the topic, and answering questions about the topic with ease.

CO –PO/PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	2	1	1	1	1	1

(1–Low2–Moderate3–High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	1	1	1	1	1
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem in real manner than theoretical explanation	Students will be able to provide solution for the problem by writing python code through this activity	As the students are involved in the activity as team of two, the team work skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	The problem solving skill earned through this activity helps the students in solving various problems

Images/Screenshot of the practice:



Fig 1. Mr. Vimal. S started to draw empty box to place the list of numbered cards in his hand.



Fig 2. Ms. Harini M tried to insert a new card in to the list of ordered card

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Through this activity the students felt easy to write algorithm and draw flowchart.

- ❖ **Benefit of the practice:**

- From this activity, the students can get more clarity in the particular topic.
 - Students can able to explain the concepts without any confusion.

- This activity brought out an interest in them and provided a happy & joyful learning environment.
- This practice has made the class more interactive than the conventional practice

Challenges faced in implementation:

- Due to time limitation only some students were asked to do presentation in front.
- All the remaining students done the same activity in their place itself.

References:

- <https://theartofeducation.edu/2022/06/jun-how-many-ways-can-you-innovate-the-way-you-teach-drawing/>
- <https://www.teachthought.com/the-future-of-learning/innovative-strategies/>

Signature of Faculty Member

HOD